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ARTICLE

Imbalanced classification of manufacturing quality conditions using cost-sensitive decision tree ensembles

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ABSTRACT
Data-driven quality control techniques are being actively developed for implementation in smart factories. Quality prediction during manufacturing processes is a good example of how big data analytics can influence advanced manufacturing environments. In this paper, the problem of classifying manufacturing process conditions into normal and defective products according to defect types is dealt with. Such a quality analysis data set is generally unbalanced because the defective rate is quite low in practice. To solve this imbalanced classification problem, a cost-sensitive decision tree ensemble algorithm is adopted to boost the small number of defective cases and assign a higher cost to the misclassification of defective products than that of normal products. C4.5 decision trees are used as base classifiers, and three cost-sensitive ensembles, AdaC1, AdaC2 and AdaC3, are tried to address the imbalanced classification. A few types of defect conditions in a real-world die-casting data set were predicted through the proposed methods. In these experiments, the cost-sensitive ensembles were able to classify the imbalanced data and detect the defect conditions more precisely and more exactly than 19 algorithms in other classification categories such as classic classifiers and ensembles, cost-sensitive single classifiers and sampling-based ensembles. Especially, the AdaC2-based method mainly outperformed all other classification algorithms in terms of performance measures such as F-measure, G-means and AUC for the die-casting quality condition classification problem.

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Imbalanced classification; manufacturing quality condition classification; decision tree ensemble; cost-sensitive ensemble classification; die-casting quality analysis

1. Introduction

Recently, many manufacturing industries have adopted information and communication technologies to improve their competitive advantage, specifically through what is known as a smart factory. A smart factory employs an advanced manufacturing environment in production and operations by embracing information technologies such as the Internet of Things (IoT), the cloud, big data analytics and cyber-physical systems. The concept of a smart factory was introduced in the Hannover Fair 2011 with the vision of Industrie 4.0 (Kagermann, Lukas, and Wahlster 2011). In the smart factories of the future, the convergence of information technology and factory automation will affect most manufacturing activities such as demand forecasting, production planning and control, scheduling, inventory and logistics management (Lueth 2016). IoT technologies gather real-time production and logistics data in the factory, the cloud servers share the data collected from the IoT and information systems via the Internet or intranet and big data analytics provide actionable knowledge and optimal solutions from the high-dimensional structured and unstructured historical data on the cloud servers for continuous improvement.

In this study, a method for production quality analysis based on manufacturing data is introduced. Several models exist for production quality analysis, such as classification models, prediction models, factor analysis and pattern discovery. Specifically, classification algorithms for imbalanced data were utilised in this paper to predict potential problematic set conditions of die-casting machines. There are several types of defects in the die-casting process, including gas porosity, cracks and informing defects. To perform the quality analysis of die-casting process, a data set was collected from new sensors installed in die-casting machines and manufacturing execution systems (MESs) in a Korean company who provides automobile parts and smart phone parts. Die-casting condition data such as injection pressure, speed, temperature and cooling time were combined to the quality inspection data of the resulted products in order to develop the production quality classification models based on historical data.

In the data set, the defect rates of the production lots in the data set were very low, ranging from nearly zero to 10%, because the data were collected from a real-world working factory. In other words, it is natural that the quantity of defective data is much smaller than that of normal data in real-life data sets. Therefore, an imbalanced classification algorithms need to be considered to develop production quality classification models.

In this research, cost-sensitive ensemble algorithms are introduced to solve the production quality classification problem. These algorithms are used to classify the set conditions of die-casting manufacturing process into defective or normal conditions for each defect type. C4.5 decision trees are embedded as the base classifiers (Quinlan 1996) and the

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AdaC1, AdaC2 and AdaC3 algorithms are used for cost-based imbalanced classification (Sun et al. 2007).

The algorithms were evaluated with the experiments on the die-casting data set, and it showed better performance than other classification algorithms. Through experiments, it was found that the implemented algorithms were able to more effectively classify the imbalanced data and detect the defect conditions of die-casting processes than other kinds of classification algorithms such as single models, ensembles and cost-sensitive single models. Especially, the AdaC2-based algorithm outperformed other classification algorithms in terms of measures such as *F*-measure, *G*-means and AUC, which are often considered as significant measures for imbalanced classification problems, although the AdaC2-based model showed slightly higher performances than sampling-based ensemble classifiers.

The proposed methods could be used for classifying the various manufacturing quality conditions according to defect types, as well as the die-casting process conditions, which were experimented for illustrating the proposed method. In particular, the AdaC2 classifier was shown to have better or similar performances compared to existing representative algorithms in most imbalanced classifier categories. Moreover, since this kind of cost-sensitive ensembles allows to adjust the sensitivities of misclassification errors per class by adjusting cost factors, the practitioners are able to build the appropriate level of quality analysis models according to their requirements.

The remainder of this paper is organised as follows. In Section 2, some related studies are described. The manufacturing quality condition classification problem is introduced and the cost-sensitive ensemble algorithms are also explained in Section 3. The experiments with a die-casting process data set are then presented in Section 4. Finally, conclusions and future work are described in Section 5.

2. Related work

Manufacturing quality analysis is generally associated with the identification and prediction of defects, as well as recommendations to prevent such defects. There are many previous studies in the manufacturing field related to quality analysis. The approaches and techniques suggested by researchers for quality analysis include the classification and root cause analysis of quality in order to reduce the number of defects and predict the quality value, thus improving the quality. These researches often use machine-learning algorithms, statistical methods and computer simulations.

Studies on reducing the defect rate include the classification of defects and the root causes analysis of defects. Das (2008) identified the root causes of the dragging problem as well as the appropriate remedial action with a view to eliminate the problem by presenting how the statistical process control technique was used to solve a quality problem through the use of statistical tools. In addition, Holst (2008) developed classifiers such as feedforward neural networks and a support vector machine to predict defects in the steel strips produced. The different kinds of feature selection methods such as a preprocessing step for the classifiers were used. Several machine-learning classifiers were compared to predict the ultimate tensile strength and microshrinkage apparition (Santos et al. 2010). Chien, Hsu and Chen (2013) developed a manufacturing intelligence solution that integrates spatial statistics and neural networks for the detection and classification of wafer bin map patterns. Moreover, a Bayesian inference-based methodology was adopted for the analysis and reduction of defects (Sata and Ravi 2017).

There have been several attempts to predict the quality values in manufacturing for various products, including surface roughness, product size, product weight and welding parameters. Most of the references that predicted the surface roughness used statistical techniques or machine-learning algorithms. Suresh, Rao and Deshmukh (2002) developed a surface roughness prediction model for machining mild steel using the response surface methodology and optimised the surface roughness prediction model using genetic algorithms. Savalina, Sabo and Šimunović (2011) analysed the effects of the cutting depth, feed rate and the number of revolutions on surface roughness, and Asiltürk et al. (2012) developed an adaptive network-based fuzzy inference system (ANFIS) for surface roughness and vibration prediction in cylindrical grinding. ANFIS and genetic algorithms were presented to model surface roughness in end milling (Samanta 2009). Akıncıoğlu et al. (2013) and Zhang et al. (2014) proposed artificial neural network-based models for surface and hole quality in the drilling of AISI D2 cold work tool steel, and a Gaussian process regression for predicting the surface roughness in end face milling in order to avoid the costly trial-and-error, respectively.

There are many studies that predicted the production quality as well as the surface roughness using different techniques. Heidarzadeh, Jabbari and Esmaily (2015) developed and applied a thermal model to simulate the friction stir welding of pure copper plates with a thickness of 2 mm to predict the grain size. Likewise, the weight of each part according to the process conditions was predicted using a genetic neural fuzzy system, which is a hybrid-learning algorithm (Li, Jia, and Yu 2004). Chen et al. (2008) presented an innovative neural network-based quality prediction system for a plastic injection moulding process using a self-organising map plus a back-propagation neural network. Some of the references presenting quality prediction methods are listed here, along with the corresponding techniques and purpose: Gaussian process regression for melt index prediction (Ge, Chen, and Song 2011), support vector machines and genetic algorithms for wafer quality prediction (Chou, Wu, and Chen 2010), a hybrid fuzzy neural network and genetic algorithm for chemical components quality prediction (Er, Liao, and Lin 2000), partial least squares for product quality prediction (Wang 2011), analysis of variance for the sink depths prediction (Mathivanan and Parthasarathy 2009), feature selection and multi-variate calibration for final quality value prediction (Cho 2015), a back-propagation neural network for pulsed metal inert gas welding prediction (Pal, Pal, and Samantaray 2010), a back-propagation neural network and corrective neural network for welding parameters prediction (Kim et al. 2003) and an artificial neural network and genetic algorithm for optimising the injection moulding process conditions (Shen, Wang, and Li 2007).

Several researchers have attempted die-casting defect analysis, which is also conducted for exemplary experiments in this paper. These studies mainly dealt with process condition optimisation. For the optimisation of process condition, the Taguchi multi-quality analytical method was adopted to find the optimal process conditions (Hsu and Do 2013; Singaram 2010), and neural networks to generate process conditions for the pressure die-casting process were used (Mohanty and Jena 2014; Yarlagadda 2000; Krimpenis et al. 2006; Singaram 2010; Zhang and Wang 2013; Karunakar and Datta 2008). Moreover, the grey-based fuzzy algorithm was adopted (Chiang, Liu, and Chou 2008) and a simulation model was used to determine the optimal process (Binu Bose and Anilkuma 2013; Fu and Yong 2009).

The related studies on quality analysis in manufacturing process mentioned above so far have been summarised in Table 1. The literatures were categorised to the types of manufacturing process such as casting, machining, moulding and joining. For each literature, the problem of quality analysis dealt with and the analysis techniques adopted were also described. As shown in Table 1, the neural networks and the genetic algorithms have often been used in many studies.

Most defect classifications or detection problems in manufacturing are considered as imbalanced data classification problems (Rokach and Maimon 2006). However, there is little published work on the analysis of manufacturing quality conditions of defects focusing on imbalanced classification. In the previous work, the AdaC2-based classification algorithm was implemented, which embraces the cost-based AdaBoost ensemble approach for classifying imbalanced data. In order to find the most suitable classification algorithm for imbalanced data, AdaC1-based and AdaC3-based algorithms were additionally implemented in this work. The implemented algorithms aim to predict manufacturing quality and solve the imbalanced classification problem. Cost-sensitive decision tree ensemble algorithms were adopted to boost the small number of defective cases and assign a higher cost to the misclassification of defective products than that of normal products. Moreover, the experiments were conducted to compare the performances of the implemented cost-sensitive ensemble algorithms and those of existing cost-sensitive single algorithms with a real-life data set obtained from an industrial die-casting foundry.

3. Manufacturing quality condition classification

3.1. Problem definition

To define the problem of manufacturing quality condition classification, the data sources related to manufacturing quality should be considered. First, field data for manufacturing conditions can be collected from production information systems such as MESs, which may also provide the raw data for the manufacturing process conditions via a variety of sensors (e.g. temperature, pressure, speed) as well as the production lot data in the production schedule. Sometimes, new sensors can be installed to capture real-time field data of the manufacturing machines and Radio-Frequency Identification tags can be affixed to items to trace them in logistics. There are too many possible combinations of field data to use all of them at once, so we must therefore choose relevant data for targeted analysis. In this paper, data related to the manufacturing process conditions and product quality inspection are mainly collected.

Since the manufacturing process conditions such as temperature, pressure and speed are gathered continuously during a certain time period (for instance, every 5 s), the raw time-series data or its representative data should be associated with a single item or an item group to analyse the influence of manufacturing process conditions on product quality. Ideally, each single item can be traced in the lot-based manufacturing process through quality inspection by identifying the exact cavity location of each item or the production sequence of the items in the production lot. Unfortunately, these two activities, i.e. the manufacturing process and quality inspection, are often separated geographically and may also have different processing units in many manufacturing fields.

For that reason, it is assumed that the manufacturing process conditions cannot be mapped exactly to each product item, though it is possible to map the conditions to a production lot associated with an item group. This type of item-to-condition mapping problem is more common for small and medium-sized manufacturers. Because there is also the difficulty of item-to-condition mapping in the experimental data set, the quality condition classification problem in this paper is designed at the level of a production lot, as opposed to a single item.

In this research, it is assumed that three kinds of data sources can be used based on the production lot, as depicted in Figure 1, and the examples of the sources are presented in Table 2. First, the process condition data include historical condition data for the manufacturing process. For example, process condition data such as injection speed and cast temperature were collected from the MES of the die-casting machines over a few months. Second, lot information data contain basic information related to production units (so-called production lots) such as the number of produced items and the start and end timestamps of each lot production. Third, lot defect data mainly contain the number of defects in each lot according to defect type. These three data sets are joined into a single data set based on timestamps and lot numbers. By using timestamps, the process condition data can be combined with lot information data, and through lot numbers, the lot information data can also be mapped to the quality inspection data. Finally, it can be induced which production conditions resulted in the defect rates of each production lot for each defect type. For example, the maximum value of the cast temperature is associated with the corresponding lot number, and the lot number is also associated with the porosity defect rate of a specific lot.

Here, the problem that will be dealt with in this paper is first defined. The main purpose of this study was to predict the potential defect types given a certain condition in the manufacturing process. This is called quality condition classification (QC2) as defined in Definition 1. Generally, it assumes that the manufacturing conditions are obtained in the form of time-series data, as shown in Table 2(a).

Table 1. Summary of literature on quality analysis in manufacturing process.

Category	Literature	Quality analysis problem	Analysis techniques
Casting	Yarlagadda (2000)	Determination of the process parameters for the pressure in die casting	NN
	Krimpenis et al. (2006)	Finding of combination of input parameters which achieves the best output parameter values in die casting	GA, NN
	Chiang, Liu, and Chou (2008)	Finding of optimal setting and the influence of parameters in die casting	Grey relational analysis, ANOVA, fuzzy
	Fu and Yong (2009)	Identification and prediction of the process-related defects in high-pressure die casting	Computer-aided engineering simulation
	Santos et al. (2010)	Prediction of ultimate tensile strength and microshrinkage apparition in casting	Bayesian networks, *k*-nearest neighbour, NN
	Singaram (2010)	Optimisation of process conditions in the pressure die-casting process	TM, statistical process control, NN
	Hsu and Do (2013)	Optimisation of parameters and factors to increase the aluminium ADC10 die-casting quality	TM
	Zhang and Wang (2013)	Optimisation of process parameters for low-pressure die casting	GA, NN
	Mohanty and Jena (2014)	Optimisation of input process parameters in die casting	ANOVA, NN, regression
	Sata and Ravi (2017).	Prediction of defects in casting	Bayesian inference
Machining	Suresh, Rao, and Deshmukh (2002)	Prediction of surface roughness for machining mild steel	Response surface methodology, GA
	Samanta (2009)	Prediction of surface roughness in end milling	Adaptive neuro-fuzzy inference, GA
	Savalina, Sabo, and Šimunović (2011)	Prediction of surface roughness for machining	Moving least squares, least absolute deviations
	Asiltürk et al. (2012)	Prediction of surface roughness and vibration in cylindrical grinding	Adaptive neuro-fuzzy inference
	Akıncıoğlu et al. (2013)	Prediction of surface and hole quality in the drilling of AISI D2 cold work tool steel	NN
	Zhang et al. (2014)	Prediction of surface roughness in end face milling	Gaussian process regression
Moulding	Li, Jia, and Yu (2004)	Prediction of weight of each part in an injection	GA, fuzzy NN
	Shen, Wang, and Li (2007)	Optimisation of the injection moulding process conditions	GA, NN
	Chen et al. (2008)	Prediction of weight for product in a plastic injection moulding	NN, self-organising map
	Mathivanan and Parthasarathy (2009)	Prediction of the sink depths in injection moulding	ANOVA
Joining	Kim et al. (2003)	Prediction of welding parameters	GA, NN
	Pal, Pal, and Samantaray (2010)	Prediction of the pulsed metal inert gas welding	NN
	Heidarzadeh, Jabbari, and Esmaily (2015)	Prediction of the grain size of the friction stir welding of pure copper plates	Thermal analysis

NN: Neural network; GA: genetic algorithms; ANOVA: analysis of variance; TM: Taguchi method.

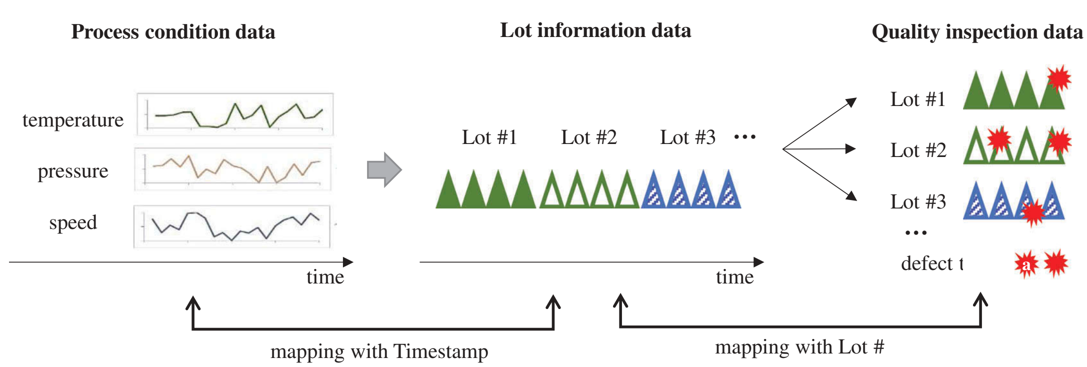


Figure 1. Mapping three data sources for manufacturing quality condition classification.

Definition 1. Quality condition classification (QC2). Given n time-series condition data $X = \{X_1, X_2, \ldots, X_n\}$ in a manufacturing process, the classification of X into a subset of defect types $D = \{D_1, D_2, \ldots, D_m\}$ is performed in such a way that the condition in the manufacturing process is mapped to a set of potential defect types based on historical data.

Herein, it was not possible to trace a single item from the manufacturing stage to the quality inspection stage. To deal with the poor item-to-condition mapping problem, the concept of lot-level quality labelling is introduced to categorise normal and defective production lots based on a given threshold. How to apply this threshold for labelling will be discussed in the next subsection.

Definition 2. Lot-level quality labelling. Lot-level quality labelling is the labelling of a lot with N defect items into a Boolean vector for defect types W_t^{C2} based on a defect criterion, θ, given the number of defect items a_t^{C2} in a lot of defect types.

Moreover, if the production lots in the data set are divided into normal and defective ones, many binary classification algorithms can be applied to the binary quality

Table 2. Example data set of manufacturing quality analysis.

	(a) Process condition data			(b) Lot information data				(c) Quality inspection data			
Machine id	Timestamp	Variable	Value	Work order id	Lot number	Start time	End time	Num. of items	Lot id	Defect type	No. of defects
B01	04–09–2015 16:00:45	Maximum speed	4.181	1150904007	1150904007-002-01	04–09–2015 16:00:42	04–09–2015 20:40:18	180	1150904007-002-01	Porosity	23
B01	04–09–2015 16:00:45	Injection pressure	723.3	1150904007	1150904007-003-01	04–09–2015 20:40:39	04–09–2015 20:51:02	180	1150904007-002-01	Trimming	35
B01	04–09–2015 16:00:45	Low-speed stroke	190.15	1150904007	1150904007-004-01	04–09–2015 20:51:23	04–09–2015 21:42:44	180	1150904007-003-01	Material	3
B01	04–09–2015 16:00:45	Clamping force	164.95	1150904007	1150904007-005-01	04–09–2015 21:43:05	04–09–2015 23:05:40	70	1150904007-003-01	Porosity	15
B01	04–09–2015 16:00:45	Low speed	0.65	1150916001	1150916001-001-01	04–09–2015 23:06:00	04–09–2015 23:57:41	180	1150904007-004-01	Imprint	6

condition classification problem, defined in Definition 3. In this paper, the BQC2 problem for several defect types will be mainly focused on. The QC2 can also be solved by applying multiple BQC2 models in parallel.

Definition 3. Binary quality condition classification (BQC2). Given a data set of n time-series manufacturing condition data $X = \{X_1, X_2, \ldots, X_n\}$, the binary classification of X into a binary value is performed for a specific defect type $B = \{-1, 1\}$ in such a way that the condition in the manufacturing process is mapped to whether the defect type may occur based on historical data. Herein, 1 indicates the potential occurrence of the defect type, while −1 represents the normal condition for the defect type.

3.2. Production data preprocessing

To prepare data for BQC2, two types of data preprocessing are introduced in this paper. The first preprocessing step is the time-series data representation of manufacturing process conditions X. Many production condition data of die-casting machines such as injection pressure, injection speed, oil temperature and die temperature are time-series data, and they cannot be used as input of typical classification algorithms such as decision trees and support vector machines. Therefore, time-series representation methods are often used to extract the features from time-series data. There are generally four categories of time-series representation methods used for data mining: model-based, data adaptive, non-data adaptive and data dictated representation approaches, as shown in Figure 2 (Aghabozorgi, Shirkhorshidi, and Wah 2015; Bagnall et al. 2006).

Most of the representation methods can be used to reduce the dimensionality and effectively represent the characteristics of the time-series. In this research, the basic statistical representatives of model-based representation approaches, such as the mean, deviation, maximum and minimum values of the time-series data, are mainly used. However, other time-series representation methods may also be used in similar ways to build models for BQC2.

The second preprocessing step is the lot-level quality labelling described in Definition 2. A production lot may contain a few types of defects simultaneously, and the defect rates generally vary according to the defect type. For instance, 180 items in the same lot may include 5 material defects and 13 porosity defects, as shown in Table 2(c). In addition, the die-casting process may have different defect rates of porosity defects and imprint defects. Therefore, assigning defect labels and defect rates into the production lots and assigning normal labels into the other lots will directly affect the algorithm performance and the utility of the quality classification models.

Two kinds of lot-level defect labelling methods were assumed in this paper according to the different defect types, as described in Definitions 4 and 5.

Definition 4. Absolute rate-based labelling (AL). Given the number of defect items in lot N_j for defect type D_j and the absolute defect rate threshold θ^A, absolute rate-based labelling (AL) considers the corresponding production lot to be a

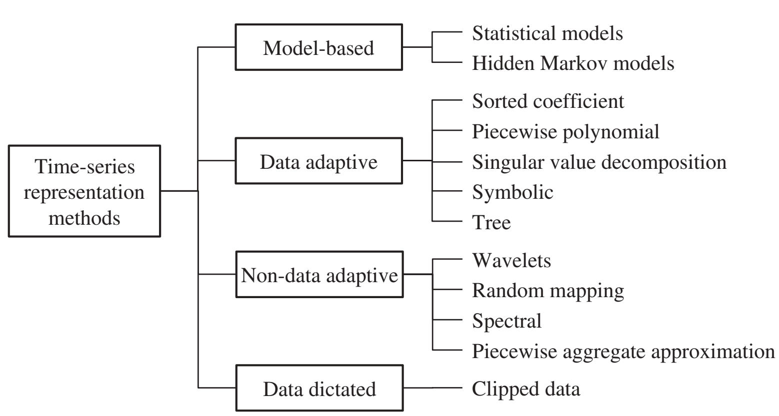


Figure 2. Categories of time-series data representation methods (Bagnall et al. 2006).

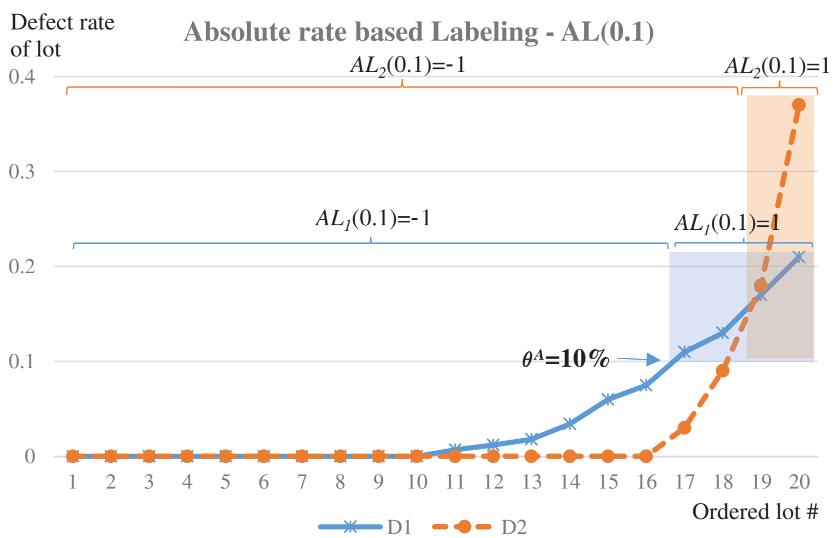
defective lot if the defect rate of the lot is greater than θ^A. That is, if $N_j/T > \theta^A$, $AL_j(\theta^A) = 1$, otherwise, $AL_j(\theta^A) = -1$, where T is the total number of items in the lot and $AL_j(\theta^A)$ is the absolute rate-based label for D_j.

Definition 5. Relative rate-based labelling (RL). Given the numbers of defect items in lot N_j for defect type D_j and the relative defect rate threshold θ^R, relative rate-based labelling (RL) considers the corresponding production lot to be a defective lot if the percentile of the defective lot among all the lots with the same defect type is greater than θ^R. That is, if $percentile(N_j, D_j) > \theta^R$, $RL_j(\theta^R) = 1$, otherwise, $RL_j(\theta^R) = -1$, where the *percentile* function maps the number of defects to the percentile rank among the lots with D_j and $RL_j(\theta^R)$ is the relative rate-based label for D_j.

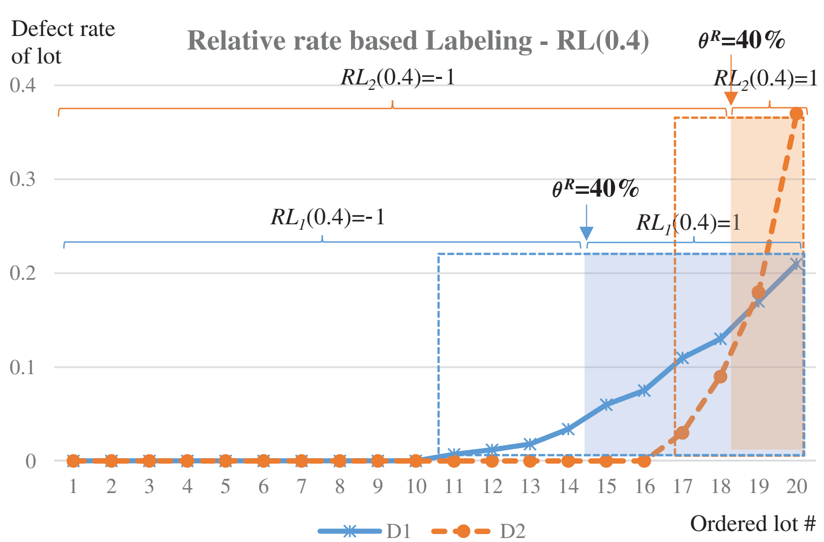
Figure 3 depicts examples of AL(0.1) and RL(0.4) for the lot defect rates of two defective types, D_1 and D_2. In Figure 3(a), AL has an absolute threshold rate of $\theta^A = 10\%$. As shown, four lots have greater defect rates than θ^A for D_1; so, they are identified as defective lots. Similarly, two lots are defective for D_2. On the contrary, in Figure 3(b), the RL with a relative threshold rate of $\theta^R = 40\%$ included six lots for D_1 and two lots for D_2 that were larger than the percentile of θ^A in the ordered lots for each defect type.

3.3 Quality condition classification algorithm

In this subsection, cost-sensitive ensemble algorithms are introduced to build the BQC2 models from a data set



(a) $AL\ (\theta^A=0.1)$



(b) $RL\ (\theta^R=0.4)$

Figure 3. Examples of absolute rate and relative rate-based labelling (AL and RL) of 20 lots for two defect types, D_1 and D_2. (a) AL with absolute threshold rate $\theta^A = 10\%$ and (b) RL with relative threshold rate $\theta^R = 40\%$.

described in Section 3.2. As discussed in the introduction, the number of defective lots is much smaller than that of the normal lots. In the example in Figure 3, more than half of the lots have zero defect rates. In this paper, cost-sensitive decision tree ensemble algorithms, called AdaC1, AdaC2 and AdaC3 (Sun et al. 2007), are adopted to solve this imbalanced BQC2 problem. These algorithms boost the small number of defective cases with the AdaBoost approach (Freund and Schapire 1995) and assign a higher cost to the misclassification of defective lots than that of normal lots. In this research, the C4.5 decision tree was used as a basic classifier (Quinlan 1996). A decision tree is one of representative prediction techniques for classification or regression, which provides conjunctive split rules from a root to leaves that conclude a class or a regression value. Recently, the decision trees are often used base classifiers of ensembles to improve the prediction performance. In this research, the three cost-sensitive decision tree ensembles are used for manufacturing quality analysis.

The AdaC1, AdaC2 and AdaC3 have the same procedure as adopted from the AdaBoost algorithm (Freund and Schapire 1995), but decide the importance of a base classifier and the sample weights in different ways. The sample weight $W_t(i)$ is the probability that the ith sample (X_i, b_i) is chosen at the tth iteration for creating a base classifier, where X_i is a tuple of attribute values belonging to a domain X and b_i is a label in a set B = {−1, 1}. In this research, X_i is prepared by extracting representative values of the time-series manufacturing condition data for the ith production lot, and b_i represents a normal or defective lot for a specific defect type (i.e. −1 and 1 indicate a normal and defective lot, respectively).

The pseudocode for the BQC2 classification algorithm based on AdaC2 is shown in Figure 4, which has been extended from Sun et al. (2007). To finally generate the ensemble classifier, the algorithm trains T decision trees for T iterations with n training data in the given manufacturing quality data set, $D = \{(X_1, b_1), \ldots, (X_n, b_n)\}$, for the BQC2 problem defined in Definition 3. A training instance (X_i, b_i) is a pair of a manufacturing condition $X_i \in X$ and a binary label for a specific defect type $B = \{-1, 1\}$. Binary labelling can be done through absolute rate-based or relative rate-based labelling, $AL_j(\theta^A)$ or $RL_j(\theta^R)$, for the jth defective type and the given threshold θ^A and θ^R, as defined in Definitions 4 and 5, respectively.

Initially, all n training instances in the original set have identical sample weights (line 1). However, for the T iterations, the updated sample weights generate different training sets for building the T base classifiers. At the tth iteration, the base decision tree classifier h_t is created by using the training set D_t, which is sampled from the original set D with the sample weights α_t^{C2} (lines 3 and 4). By applying the classifier h_t to the original training set D, it calculates the importance of the tth classifier, denoted by α_t^{C2} (lines 5 and 6). The importance of a classifier, α_t^{C2}, is measured by the logarithm of the ratio of the

input: original training set $D = \{(X_1, b_1), \ldots, (X_n, b_n)\}$, where $X_i \in X$, $b_i \in B = \{-1,1\}$

output: ensemble classifier $H(X_k)$ for $X_k \in X$

1: Initialize sample weights $W_1^{C2}(i) = 1/n$ for all n instances.

2: **for** $t = 1$ to T **do**

3: Create training set D_t by sampling from D according to W_t^{C2}.

4: Train a base classifier h_t on D_t.

5: Apply h_t to all instances in D.

6: Calculate the importance of h_t:

$$\alpha_t^{C2} = \frac{1}{2}\log\frac{\sum_{i,b_i=h_t(X_i)} C_i W_t^{C2}(i)}{\sum_{i,b_i\neq h_t(X_i)} C_i W_t^{C2}(i)} \tag{1}$$

7: **if** $\alpha_t^{C2} < 0$ **then**

8: Reset sample weights $W_t^{C2}(i) = 1/n$ for all n instances.

9: Go back to Step 3.

10: **end if**

11: Update the sample weights for all n instances:

$$W_{t+1}^{C2}(i) = \frac{C_i W_t^{C2}(i)\exp(-\alpha_t^{C2} h_t(X_i) b_i)}{\sum_i C_i W_t^{C2}(i)\exp(-\alpha_t^{C2} h_t(X_i) b_i)} \tag{2}$$

12: **end for**

13: Create an ensemble classifier:

$$H(X_k) = sign(\sum_{t=1}^{T} \alpha_t^{C2} h_t(X_k)) \tag{3}$$

Figure 4. AdaC2-based BQC2 classifier algorithm.

corrected classification costs to the misclassification costs, as described in Equation (1) in Figure 4. If W_{t+1}^{C2} is negative, it cannot improve the misclassification. Therefore, the algorithm resets the sample weights and then restarts to create a training set (lines 8 and 9). Then, $\alpha_t^{C1} = \frac{1}{2}\log\frac{1+\sum_{i,b_i=h_t(X_i)} C_i W_t^{C1}(i) - \sum_{i,b_i\neq h_t(X_i)} C_i W_t^{C1}(i)}{1-\sum_{i,b_i=h_t(X_i)} C_i W_t^{C1}(i) + \sum_{i,b_i\neq h_t(X_i)} C_i W_t^{C1}(i)}$ is used as the weight of the tth classifier h_t when voting at the final classifier and is also utilised in updating the sample weights $W_{t+1}^{C1}(i) = \frac{W_t^{C1}(i)\exp(-\alpha_t^{C1} C_i h_t(X_i) b_i)}{\sum_i W_t^{C1}(i)\exp(-\alpha_t^{C1} C_i h_t(X_i) b_i)}$ at the next iteration. The sample weights are updated according to the classification result of h_t for all n training instances, as shown in Equation (2) in Figure 4. Finally, after T iterations, the algorithm obtains the ensemble classifier, which can classify a new instance based on the weighted votes from the T base classifiers, as shown in Equation (3) in Figure 4.

The algorithm can adopt one of three cost-sensitive algorithms, AdaC1, AdaC2 or AdaC3, by choosing the equations for the classifier importance and the sample weights, as summarised in Table 3 (Sun et al. 2007). The classifier importance α_t plays the role of decreasing the sample weights of correctly classified instances and increasing the sample weights of the misclassified instances at the next iteration of W_{t+1}.

4. Experiments

4.1 Die-casting case

Die casting is one of the most widely used manufacturing processes for production parts (Syrcos 2003). The die-casting process conditions heavily affect the quality of the products. For BQC2 experiments, a real-life die-casting data set was obtained from a company that supplied various magnesium, aluminium and zinc parts to automobile part makers and major mobile phone companies in Korea. The historical data including processing conditions and quality inspections were collected from new sensors installed at die-casting machines and MESs.

The die-casting process and the related data are depicted in Figure 5. The processing condition variables include high speed, high-speed time, cooling time, injection pressure, biscuit thickness, die temperature, oil temperature, low speed, low-speed stroke, low-speed time, casting pressure, maximum speed and clamping force. The production is managed in the production lot unit and is inspected individually through sampling. The defect types of the die-casting process are gas porosity, imprint, trimming, informing defect, shrinkage porosity, hot tears, flow marks and so forth. In the experiments, whether these defects occur in a given production lot will be predicted based on the values of the die-casting process conditions.

4.2 Data set

In this case study, three kinds of data were collected from the MESs and the newly installed sensors in the die-casting factory: process condition data, lot information data and lot defect data. These data were joined and prepared as training and testing data for constructing BQC2 models. A sample of the joined data is shown in Table 4, which contains process conditions, basic lot information and the binary labelling for porosity defects. For the construction of the BQC2 models, several representative values, such as mean, maximum, minimum, standard deviation and the numbers of instances deviating from the upper and lower limits, were first extracted from each time-series process condition variables for each production lot. In Table 4, only the mean values of a few condition variables like maximum speed, injection pressure and biscuit thickness were presented because of the limited space.

From the lot size and the number of porosity defects of a given lot, defect or normal labels were assigned to the lot (1 for defective lots and −1 for normal lots) by using Definitions 4 and 5. For instance, in the case of $AL_p(0)$ labelling, which is an absolute rate-based labelling for porosity defects with an absolute defect rate threshold of $\theta^A = 0$, the lots with any number of porosity defects have $AL_p(0) = 1$ and the others have $AL_p(0) = -1$. In the case of $AL_p(0.1)$ labelling, the lots with porosity defect rates higher than 10% have $AL_p(0.1) = 1$, while the others have $AL_p(0.1) = -1$. For $RL_p(0.4)$ labelling, only the lots with 60% of the highest value of defective lots among all lots with porosity defects have $RL_p(0.4) = 1$, and the others have $RL_p(0.4) = -1$. This labelling is critical to the performance of BQC2 models since it is used as a criterion when training and determining defective lots. In the same way, defect or normal labels were assigned to each lot for other defects types such as trimming, material and imprint defects in this experimental study.

4.3 Experimental design

There are classic single classifiers such as C4.5, k-NN (k-nearest neighbours), CART (classification and regression tree) and SVM

Table 3. Equations of classifier importances and sample weights for AdaC1, AdaC2, and AdaC3.

Algorithm	Importance of classifier at iteration t	Sample weight for the i-th instance at iteration t+1
AdaC1	$\alpha_t^{C1} = \frac{1}{2}\log\frac{1+\sum_{i,b_i=h_t(X_i)} C_i W_t^{C1}(i) - \sum_{i,b_i\neq h_t(X_i)} C_i W_t^{C1}(i)}{1-\sum_{i,b_i=h_t(X_i)} C_i W_t^{C1}(i) + \sum_{i,b_i\neq h_t(X_i)} C_i W_t^{C1}(i)}$	$W_{t+1}^{C1}(i) = \frac{W_t^{C1}(i)\exp(-\alpha_t^{C1} C_i h_t(X_i) b_i)}{\sum_i W_t^{C1}(i)\exp(-\alpha_t^{C1} C_i h_t(X_i) b_i)}$
AdaC2	$\alpha_t^{C2} = \frac{1}{2}\log\frac{\sum_{i,b_i=h_t(X_i)} C_i W_t^{C2}(i)}{\sum_{i,b_i\neq h_t(X_i)} C_i W_t^{C2}(i)}$	$W_{t+1}^{C2}(i) = \frac{C_i W_t^{C2}(i)\exp(-\alpha_t^{C2} h_t(X_i) b_i)}{\sum_i C_i W_t^{C2}(i)\exp(-\alpha_t^{C2} h_t(X_i) b_i)}$
AdaC3	$\alpha_t^{C3} = \frac{1}{2}\log\frac{\sum_i C_i W_t^{C3}(i) + \sum_{i,b_i=h_t(X_i)} C_i^2 W_t^{C3}(i) - \sum_{i,b_i\neq h_t(X_i)} C_i^2 W_t^{C3}(i)}{\sum_i C_i W_t^{C3}(i) - \sum_{i,b_i=h_t(X_i)} C_i^2 W_t^{C3}(i) + \sum_{i,b_i\neq h_t(X_i)} C_i^2 W_t^{C3}(i)}$	$W_{t+1}^{C3}(i) = \frac{C_i W_t^{C3}(i)\exp(-\alpha_t^{C3} C_i h_t(X_i) b_i)}{\sum_i C_i W_t^{C3}(i)\exp(-\alpha_t^{C3} C_i h_t(X_i) b_i)}$

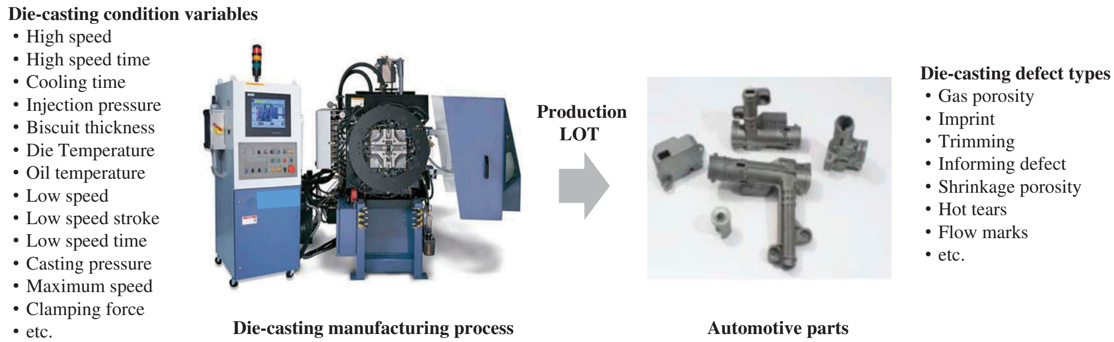


Figure 5. Example of die-casting process with condition variables and defect types.

Table 4. Sample data of lot-based die-casting process conditions, quality inspection data (for porosity defects) and binary labelling.

	Representation of process conditions (mean, max, min, sd, UL, LL)						Lot information		Binary labelling		
Lot number	Maximum speed (μ)	Injection pressure (μ)	Biscuit thickness (μ)	Low-speed stroke (μ)	Clamping force (μ)	Low speed (μ)	Lot size	Number of porosity defects	$AL_p(0)$	$AL_p(0.1)$	$RL_p(0.4)$
1150904007-002-01	4.40	728.93	18.23	190.13	165.07	0.66	180	23	1	1	1
1150904007-003-01	4.59	732.64	16.05	190.12	161.72	0.66	180	0	−1	−1	−1
1150904007-004-01	3.69	1212.48	9.62	190.77	159.08	0.76	180	15	1	−1	1
1150904007-005-01	4.91	737.83	14.02	192.12	151.11	0.65	180	0	−1	−1	−1
1150916001-001-01	4.41	732.60	14.97	190.12	165.72	0.66	180	7	1	−1	−1

(support vector machine), which can be used for BQC2 problems. However, it has been well known that ensemble algorithms such as Boosted tree, Random forest and AdaBoost often outperform the classic single classifiers in many applications because the ensembles of classifiers can complement with one another the weakness of their individual classifiers.

In the meantime, since many real-world classification applications have the class imbalance problem, various imbalanced algorithms have been proposed to solve this problem (Japkowicz and Stephen 2002). The common approaches to the imbalanced classifications are sampling methods and cost-sensitive methods (Haibo and Garcia 2009), and the ensembles combined to the two approaches are often used to solve the problem. Galar et al. (2012) divided the ensemble algorithms for the imbalanced classification into two categories, sampling-based algorithms (sampling + ensembles) and cost-sensitive ensembles. They subdivided the first category again to cost-sensitive boosting, boosting-based, bagging-based and hybrid ensembles, and they showed that RUSBoost, SMOTEBagging and UnderBagging had the best performances by several experiments.

In this paper, the performance of the proposed BQC2 methods, AdaC1, AdaC2 and AdaC3, will be compared with those of 19 algorithms in 4 other categories: classic single classifiers (C4.5, *k*-NN, CART, linear SVM and radial SVM), classic ensemble algorithms (boosted tree, random forest, bagged CART, AdaBoost and AdaBoost.M1), cost-sensitive single classifiers (cost-sensitive CART, cost-sensitive linear SVM, cost-sensitive L2 SVM and cost-sensitive radial SVM) and sampling + ensembles (RUSBoost, SMOTEBagging, UnderBagging, EasyEnsemble and BalanceCascade). Table 5 provides the list of total 22 classifiers with their short descriptions and their tuning parameters. The AdaC1, AdaC2 and AdaC3 algorithms were implemented in Java, while the other algorithms were executed using R packages and Python packages.

In many studies on empirical comparison of algorithms, the parameters recommended by the inventors or the default parameters in their packages are used to build a model (Liu, Wu, and Zhou 2009; Seiffert et al. 2010; Galar et al. 2012). However, the algorithms cannot achieve their best results without fine-turning their parameters depending on the data set. Therefore, model selection steps including parameter tuning should be included in the resampling, i.e. repetition for every pair of training and test data in order to estimate properly the performance of algorithms. In this context, a nested cross-validation was adopted for the experiments as a convincible resampling technique for model evaluation with parameter tuning and algorithm selection. In the nested cross-validation procedure, an inner cross-validation is first used to tune the parameters and select the best model. In this step, the most common and reliable approach to parameter selection is to decide on the set of candidate values of parameters and then choose the optimal settings (Staelin 2003). An outer cross-validation is then executed to evaluate the model selected by the inner cross-validation. In these experiments, the resampling methods mentioned above with 5-fold cross-validation in the inner loop for parameter tuning, and 5-fold cross-validation in the outer loop for model evaluation. So,

Table 5. Summary of BQC2 classifiers adopted for experiments.

Classifier		Short description	Parameters
Classic single classifiers	C4.5	C4.5 decision tree learning algorithm (Quinlan 1996)	Confidence threshold (cf) Minimum instances per leaf (*m*)
	k-NN	*k*-Nearest neighbours learning algorithm (Ripley 2007)	Number of neighbours (*k*)
	CART	CART learning algorithm (Breiman et al. 1984)	Complexity (cp)
	Linear SVM	SVM learning algorithm with linear kernel (Cortes and Vapnik 1995)	Cost (*c*)
	Radial SVM	SVM learning algorithm with radial basis function kernel (Cortes and Vapnik 1995)	Sigma ($\mu \pm \sigma$) Cost (*c*)
Classic ensembles	Boosted tree	Gradient descent boosting using trees (Breiman 1998, Breiman et al. 1984)	Number of boosting iterations (*i*) Maximum tree depth (*d*) Shrinkage (nu)
	Random forest	Classification and regression based on a forest of trees using random inputs (Breiman 2001)	Number of randomly selected predictors (*n*)
	Bagged CART	Bagging with CART (Breiman 1996, Breiman et al. 1984)	N/A
	AdaBoost	Adaptive boosting learning algorithm (Freund and Schapire 1995)	Number of trees (*n*)
	AdaBoost.M1	Multi-class AdaBoost, slightly different weight update with AdaBoost (Schapire and Singer 1999)	Number of trees (*n*) Maximum tree depth (*d*) Coefficient type (coef)
Cost-sensitive single classifiers	Cost-sensitive CART	CART with class weights (Breiman et al. 1984)	Complexity (cp) Cost (*c*)
	Cost-sensitive Linear SVM	Linear SVM with class weights (Cortes and Vapnik 1995)	Class weight (*w*) Cost (*c*)
	Cost-sensitive L2-SVM	L2 regularised Linear SVM with class weights (Fan et al. 2008)	Loss function (*l*) Class weight (*w*) Cost (*c*)
	Cost-sensitive radial SVM	Radial SVM with class weights (Cortes and Vapnik 1995)	Sigma ($\mu \pm \sigma$) Class weight (*w*) Cost (*c*)
Sampling + ensembles	RUSBoost	AdaBoost.M1 with random undersampling in each iteration (Seiffert et al. 2010)	Sampling percentage (sp) Number of classifiers (*n*) Weight updating method (coef)
	SMOTEBagging	Bagging with each bag's Synthetic Minority Over-sampling TEchnique (Wang and Yao 2009)	Resampling ratio (*r*) Number of classifiers (*n*) Number of neighbours (*k*)
	UnderBagging	Bagging with random undersampling (Barandela, Valdovinos, and Sánchez 2003)	Resampling ratio (*r*) Number of classifiers (*n*)
	EasyEnsemble	UnderBagging but learning each bag with AdaBoost (Liu, Wu, and Zhou 2009)	Number of subsets (*s*)
	BalanceCascade	Similar to EasyEnsemble but removing examples from the majority class in each bagging iteration (Liu, Wu, and Zhou 2009)	Number of maximum subsets (ms)
Cost-sensitive ensembles	AdaC1	AdaBoost with costs inside the exponent (Sun et al. 2007)	Cost factor (*C*)
	AdaC2	AdaBoost with costs outside the exponent (Sun et al. 2007)	Cost factor (*C*)
	AdaC3	AdaBoost with costs both inside and outside exponents (Sun et al. 2007)	Cost factor (*C*)

CART: Classification and regression tree; SVM: support vector machines.

there are five pairs of training and test set in the outer loop based on stratified random sampling. On each outer training set, optimal parameters are determined in the given set of candidate values of parameters to maximise the value of the area under the receiver operating characteristic curve (AUC) by executing the inner loop, since the AUC is often used to evaluate the imbalanced classification performance. Subsequently, the classifiers are fitted on each outer training set using the corresponding selected parameters and their performances are evaluated on the outer test sets.

For evaluating performances of classification algorithms, accuracy, recall, precision and *F*-measure are commonly used. However, for evaluating overall performances of imbalanced classification problems, geometric mean (*G*-mean) and receiver operating characteristic curve (ROC) analysis are better choices (Wang and Yao 2009). *G*-mean is the geometric average of recall values of each class and ROC is the curved graph with the true positive rate on the *y*-axis versus the false-positive rate on the *x*-axis (Bradley 1997). For the ROC analysis, AUC can be used to provide a single measure for comparing performances of models. This paper used the AUC as a core performance measure to select tuning parameters and evaluate the models. Depending on the selection of the evaluation measures, different results will be observed. This paper provides four common performance measures often used for experimental results (accuracy, recall, precision and *F*-measure), and two performance measures used when data are imbalanced (*G*-mean and balanced accuracy).

4.4 Experimental results

This subsection describes the performances of BQC2 models including the proposed methods for the die-casting process

Table 6. Performance results of BQC2 classifiers for porosity defects.

Category	Classifier	Selected parameters	Accuracy	Balanced accuracy	Precision	Recall	*F*-measure	*G*-mean (σ)	AUC (σ)
Classic single classifiers	C4.5	cf = 0.255, m = 3	0.9065	0.7001	0.4643	0.3250	0.3824	0.5595 ± 0.1557	0.6442 ± 0.0668
	k-NN	k = 5	0.9120	0.7256	0.5077	0.4125	**0.4552**	**0.6296 ± 0.0584**	**0.6867 ± 0.0360**
	CART	cp = 0.03125	0.9076	0.6805	0.4400	0.1375	0.2095	0.3676 ± 0.2812	0.5602 ± 0.0744
	LinearSVM	c = 1	0.9098	0.6225	0.3333	0.0125	0.0241	0.1117 ± 0.1000	0.5050 ± 0.0132
	RadialSVM	$\mu \pm \sigma$=0.0001479, c = 0.5	0.9053	0.5930	0.2727	0.0375	0.0659	0.1927 ± 0.1220	0.5139 ± 0.0170
Classic ensembles	Boosted tree	i = 100, d = 2, nu = 0.1	0.8998	0.6646	0.4000	0.2500	0.3077	0.4907 ± 0.1905	0.6067 ± 0.0889
	Random forest	n = 61	0.9042	0.5561	0.2000	0.0250	0.0444	0.1573 ± 0.1217	0.5076 ± 0.0146
	Bagged CART	–	0.8931	0.6670	0.3947	0.3750	0.3846	0.5949 ± 0.0277	0.6594 ± 0.0157
	AdaBoost	n = 50	0.9042	0.6960	0.4531	0.3625	0.4028	0.5891 ± 0.0210	0.6599 ± 0.0132
	AdaBoost.M1	n = 50, d = 2, coef = Freund	0.9109	0.7234	0.5000	0.4500	**0.4737**	**0.6559 ± 0.0488**	**0.7030 ± 0.0313**
Cost-sensitive single classifiers	Cost-sensitive CART	cp = 0.03125, c = 3	0.9076	0.6390	0.3636	0.0500	0.0879	0.2226 ± 0.1957	0.5207 ± 0.0415
	Cost-sensitive Linear SVM	w = 2, c = 0.25	0.8942	0.7038	0.4400	0.6875	0.5366	0.7929 ± 0.0789	0.8010 ± 0.0672
	Cost-sensitive L2-SVM	l = L2, w = 3, c = 0.25	0.9076	0.5558	0.2000	0.0125	0.0235	0.1115 ± 0.0997	0.5038 ± 0.0127
	Cost-sensitive radial SVM	$\mu \pm \sigma$=0.0001055, w = 1.2, c = 0.55	0.8964	0.7125	0.4511	0.7500	**0.5634**	**0.8265 ± 0.0358**	**0.8304 ± 0.0336**
Sampling + ensembles	RUSBoost	sp = 0.1, n = 150, coef = Breiman	0.8931	0.7225	0.4518	0.9375	**0.6098**	**0.9128 ± 0.0212**	0.9131 ± 0.0213
	SMOTEBagging	r = 1:1, n = 30, k = 15	0.9165	0.7473	0.5225	0.7250	0.6073	0.8234 ± 0.0727	0.8301 ± 0.0662
	UnderBagging	r = 1:1, n = 40	0.8831	0.7119	0.4294	0.9500	0.5914	0.9125 ± 0.0203	**0.9133 ± 0.0207**
	EasyEnsemble	s = 20	0.8608	0.6875	0.3834	0.9250	0.5421	0.8891 ± 0.0267	0.8898 ± 0.0272
	BalanceCascade	ms = 40	0.8653	0.6864	0.3867	0.8750	0.5364	0.8696 ± 0.0334	0.8697 ± 0.0343
Cost-sensitive ensembles	AdaC1	C = 0.7	0.8963	0.7010	0.4414	0.6125	0.5131	0.7523 ± 0.1022	0.7683 ± 0.0762
	AdaC2	C = 0.7	0.8896	0.7195	0.4444	0.9500	**0.6056**	**0.9163 ± 0.0210**	**0.9169 ± 0.0214**
	AdaC3	C = 0.8	0.8785	0.7037	0.4171	0.9125	0.5725	0.8936 ± 0.0667	0.8938 ± 0.0619

Bold values mean the best result in each category.

data set that contain the lot-level AL(0) labelling for four types of die-casting defects.

First, the performance results of 22 BQC2 models in five classifier categories for the porosity defect type were summarised as shown in Table 6. Among the parameters selected by inner fivefold cross-validation for each outer training set, only the most frequently selected parameters for the outer test data were presented in the table due to the limited space.

To show the performances of the 22 BQC2 classifiers, seven performance measures were used. Accuracy is the rate of corrected classified instances among total instances, while balanced accuracy is the mean of the majority accuracy and the minority accuracy, i.e. the mean of the accuracy of normal lots and the accuracy of defective lots. Precision is the rate of the correctly classified defective lots among the lots that were classified into defective lots, while recall is the rate of the correctly classified defective lots among the actual defective lots. *F*-measure, also known as F1-score, is the harmonic mean of precision and recall. *G*-mean is the geometric average of recall values of normal and defective classes, and AUC is the area under ROC curve for the defective class. The higher each measure is, the better the performance is. Especially, balanced accuracy, *G*-mean and AUC are often used to evaluate the results of the imbalanced classification, because they consider the significance of the minority classification similarly weighted to that of the majority classification. Nevertheless, *F*-measure is also significant since it can show the overall performance of classification regardless of normal and defect labels. Therefore, *F*-measure, *G*-mean and AUC will be used as main evaluation measures for this BQC2 experiment. The values of seven performance measures for all classifiers in the five categories can also be compared in a chart of Figure 6.

As shown in the figure, most classifiers showed high accuracies of around 0.9, because the porosity defect data set had unbalanced classes. The degree of class imbalance is often measured with the imbalanced ratio (IR) (Chawla, Japkowicz, and Kotcz 2004). The IR of a binary data set is the ratio of the number of majority instances to that of minority instances. The porosity defect data set had IR of 10.23, which means that normal lots were about 10 times of defective lots in the data set. For balanced accuracy, many of classic single classifiers and ensembles showed the values lower than 0.7. If the classifier has the accuracy of 0.9 and the balanced accuracy of 0.7 for an imbalanced data set, it is expected that the classifier has approximately the defective accuracy of around 0.5. Therefore, some of classic single classifiers and ensembles were not reliable to the classification of porosity defective lots.

In case of *F*-measure, balancing precision and recall, all of classic single classifiers and classic ensembles had the values lower than 0.5, while all of sampling + ensembles and cost-sensitive ensembles had the values higher than 0.5. Especially, RUSBoost, SMOTEBagging, UnderBagging and AdaC2 showed the *F*-measures of around 0.7. For *G*-mean and AUC, the classifiers in the first two categories had nearly the values lower than 0.7, while most of sampling + ensembles and cost-sensitive ensembles had the values of around 0.9. Based on the three main measures, it can be said that three ensemble classifiers, RUSBoost, UnderBagging and AdaC2, showed overall the best performances. In particular, AdaC2, one of the proposed BQC2 methods, had the best performances in terms of *G*-mean and AUC, although it had slightly lower *F*-measure than RUSBoost and UnderBagging classifiers. This was consistent with the experimental results that RUSBoost and

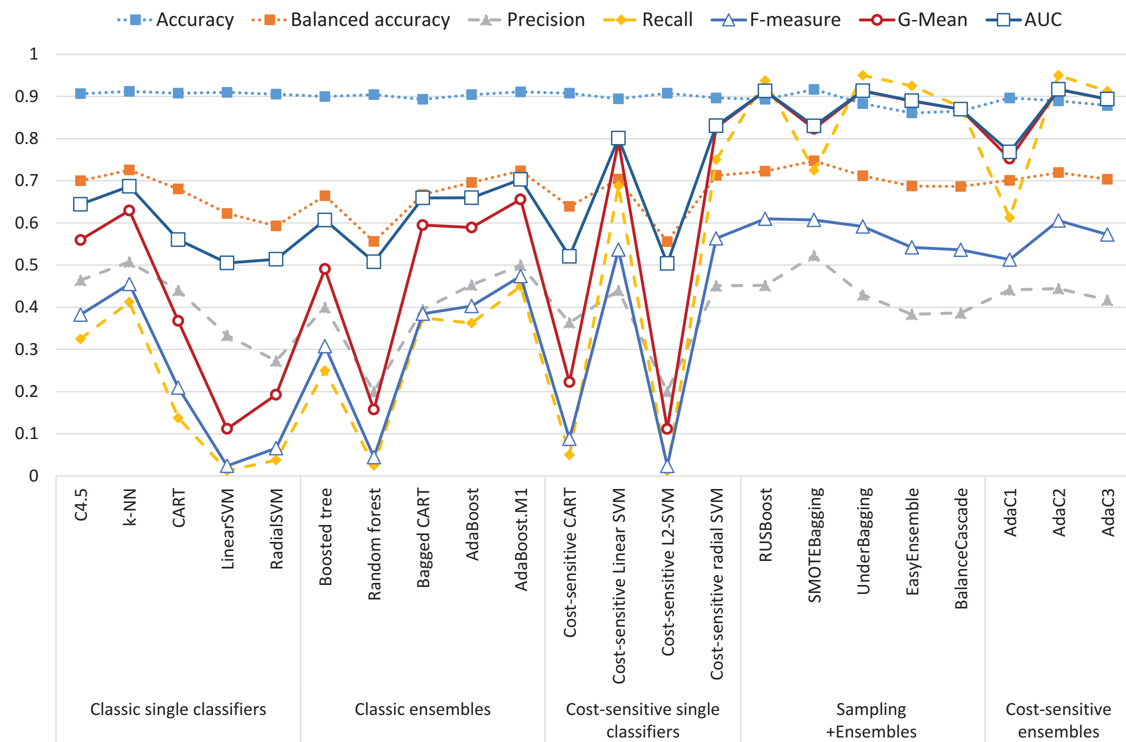


Figure 6. Performance comparison among 22 BQC2 methods in five classification categories for porosity defects.

UnderBagging achieved higher performances than other sampling-based ensembles for class imbalance problems reported in Seiffert et al. (2010), Galar et al. (2012).

In addition, experiments were conducted using data sets of three other defect types: trimming, material and imprint defects. The three data sets had the IRs of 10.23, 9.09, 15.33 and 1.41, respectively. Note that the imprint defect data set was not imbalanced data because IR was nearly one. It is because many lots of die-casting process often had small numbers of imprint defective products and the lots with any defective products were considered defective lots (i.e. AL(0)). In these experiments, the eight BQC2 models in only the best two categories, sampling + ensembles and cost-sensitive ensembles, were evaluated since they outperformed the other categories in the experiments for porosity defect data. Table 7 shows test performance results with tuning parameter selected for the three defect types, and Figure 7 provides the performance comparison of the classifiers for each defect type.

For trimming defects shown in Table 7(a), RUSBoost and UnderBagging were two best classifiers in terms of G-mean and AUC in the sampling + ensemble category. And, the AdaC2 showed slightly higher performance than the two and was the best model among all the classifiers. For material defects shown in Table 6(b), somewhat different results were observed. UnderBagging showed the best performance in its category and outperformed AdaC2, which were the best in the cost-sensitive ensembles. Finally, Table 7(c) shows the performances of the 8 BQC2 classifiers for imprint defects. See the values of accuracies, balanced accuracies, precisions and recalls. Most performance values were nearly 0.9 and it means that the imprint defect data set was quite easy to classify and moreover, it was not an imbalanced data. AdaC3 had the best performance, but the difference was too slight compared to the performances of AdaC1, AdaC2 and RUSBoost.

Additionally, the sensitivity of cost factor *C* was investigated for the AdaC2 method, which had overall the best performances for all die-casting defect types. Figure 8 shows the changes of AdaC2's performances according to cost factor *C*. Note $C = 1.0$ means not to consider class imbalance because the algorithm weighs the misclassification cost of defective lots the same as that of normal lots.

From the first three charts, it was be found that the decrease of *C* from 1.0 to 0.9 or 0.8 could improve the performances of *F*-measure, *G*-mean and AUC dramatically, because the first three data sets were imbalanced data (IR = 10.23, 9.09, and 15.33, respectively). However, in the last chart for imprint defect data set (IR = 1.41), the decrease of *C* from 1.0 hardly improved the performances and moreover, the decrease to lower than 0.5 degraded the performance seriously.

The strength of cost-sensitive ensemble classifiers is that they allow to adjust their cost parameters according to the sensitivity of misclassification errors per class. In particular, AdaC1, AdaC2 and AdaC3 enable to set cost factor *C* as the ratio of misclassification cost of normal lots to that of defective lots. After drawing the chart of performance changes according to the cost factor, the practitioner can choose the proper value based on the seriousness of the defects to construct the best BQC2 models.

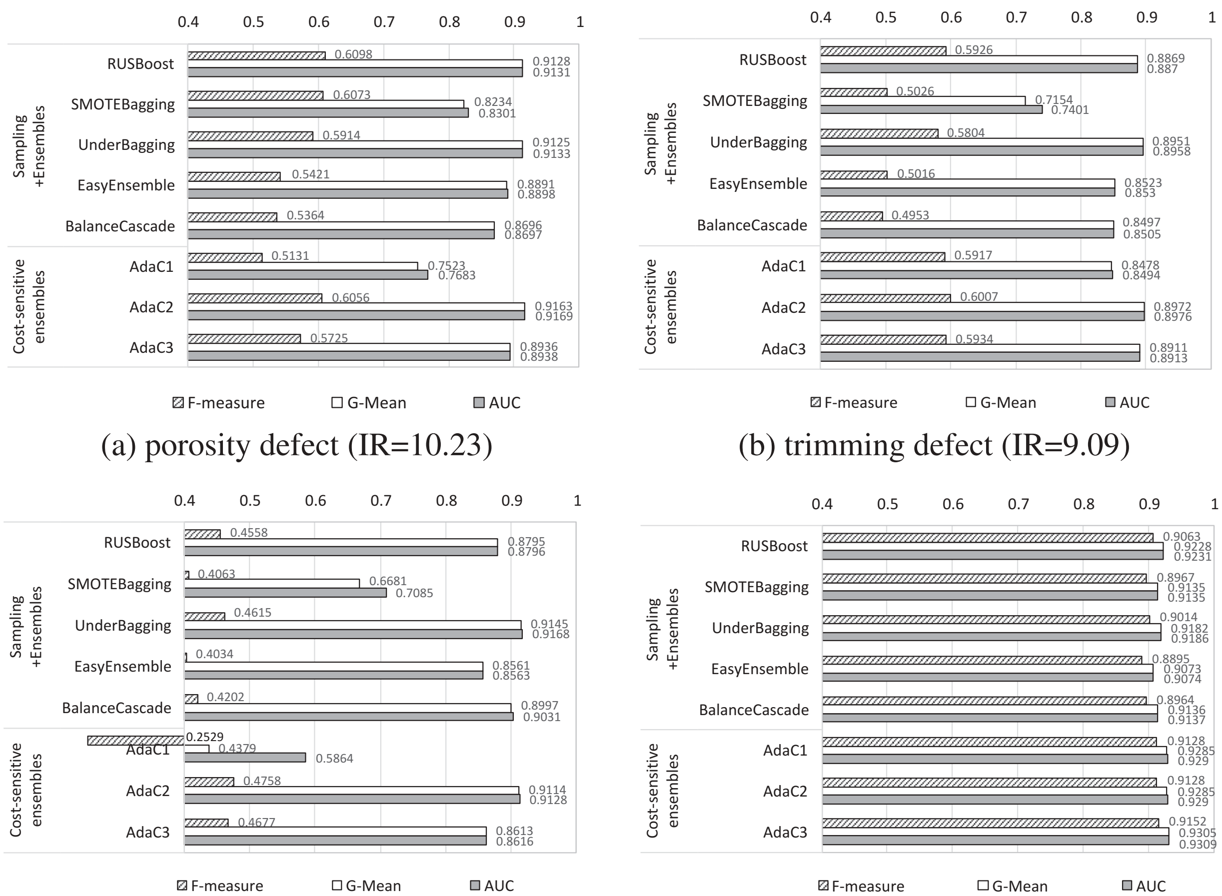
5. Conclusions and future work

In this research, the binary quality conditions classification (BQC2) problem was dealt with at the level of production

Table 7. Performance results of BQC2 classifiers for trimming, imprint and material defects.

Category	Classifier	Selected parameters	Accuracy	Balanced accuracy	Precision	Recall	*F*-measure	*G*-mean ($\mu \pm \sigma$)	AUC ($\mu \pm \sigma$)
(a) Trimming defect									
Sampling + ensembles	RUSBoost	sp = 0.1, n = 150, coef = Breiman	0.8775	0.7147	0.4420	0.8989	**0.5926**	0.8869 ± 0.0315	0.8870 ± 0.0322
	SMOTEBagging	r = 1:1, n = 20, k = 10	0.8920	0.7059	0.4623	0.5506	0.5026	0.7154 ± 0.0965	0.7401 ± 0.0716
	UnderBagging	r = 1:1, n = 20	0.8664	0.7064	0.4213	0.9326	0.5804	**0.8951 ± 0.0298**	**0.8958 ± 0.0306**
	EasyEnsemble	s = 20	0.8252	0.6673	0.3496	0.8876	0.5016	0.8523 ± 0.0460	0.8530 ± 0.0479
	BalanceCascade	ms = 30	0.8207	0.6643	0.3435	0.8876	0.4953	0.8497 ± 0.0466	0.8505 ± 0.0474
Cost-sensitive ensembles	AdaC1	C = 0.2	0.8907	0.7230	0.4702	0.7978	0.5917	0.8478 ± 0.0246	0.8494 ± 0.0232
	AdaC2	C = 0.5	0.8785	0.7179	0.4457	0.9213	**0.6007**	**0.8972 ± 0.0252**	**0.8976 ± 0.0255**
	AdaC3	C = 0.8	0.8763	0.7145	0.4402	0.9101	0.5934	0.8911 ± 0.0270	0.8913 ± 0.0277
(b) Material defect									
Sampling + ensembles	RUSBoost	sp = 0.1, n = 150, coef = Zhu	0.8697	0.6491	0.3063	0.8909	0.4558	0.8795 ± 0.0342	0.8796 ± 0.0331
	SMOTEBagging	r = 1:1, n = 30, k = 10	0.9154	0.6605	0.3562	0.4727	0.4063	0.6681 ± 0.0871	0.7085 ± 0.0574
	UnderBagging	r = 1:1, n = 50	0.8597	0.6501	0.3017	0.9818	**0.4615**	**0.9145 ± 0.0198**	**0.9168 ± 0.0203**
	EasyEnsemble	s = 20	0.8419	0.6263	0.2623	0.8727	0.4034	0.8561 ± 0.0420	0.8563 ± 0.0423
	BalanceCascade	ms = 60	0.8341	0.6329	0.2673	0.9818	0.4202	0.8997 ± 0.0268	0.9031 ± 0.0272
Cost-sensitive ensembles	AdaC1	C = 0.7	0.9278	0.6515	0.3548	0.1964	0.2529	0.4379 ± 0.2026	0.5864 ± 0.0536
	AdaC2	C = 0.8	0.8678	0.6565	0.3158	0.9643	**0.4758**	**0.9114 ± 0.0212**	**0.9128 ± 0.0221**
	AdaC3	C = 0.8	0.8811	0.6561	0.3241	0.8393	0.4677	0.8613 ± 0.0317	0.8616 ± 0.0304
(c) Imprint defect									
Sampling + ensembles	RUSBoost	sp = 0.9, n = 150, coef = Breiman	0.9187	0.9143	0.8673	0.9489	**0.9063**	**0.9228 ± 0.0088**	**0.9231 ± 0.0088**
	SMOTEBagging	r = 1:1, n = 30, k = 15	0.9120	0.9077	0.8728	0.9220	0.8967	0.9135 ± 0.0125	0.9135 ± 0.0125
	UnderBagging	r = 1:1, n = 40	0.9140	0.9097	0.8606	0.9462	0.9014	0.9182 ± 0.0125	0.9186 ± 0.0126
	EasyEnsemble	s = 30	0.9053	0.9008	0.8615	0.9194	0.8895	0.9073 ± 0.0160	0.9074 ± 0.0160
	BalanceCascade	ms = 50	0.9109	0.9064	0.8650	0.9301	0.8964	0.9136 ± 0.0172	0.9137 ± 0.0170
Cost-sensitive ensembles	AdaC1	C = 0.9	0.9242	0.9199	0.8725	0.9570	**0.9128**	**0.9285 ± 0.0160**	**0.9290 ± 0.0159**
	AdaC2	C = 0.9	0.9242	0.9199	0.8725	0.9570	**0.9128**	**0.9285 ± 0.0184**	**0.9290 ± 0.0183**
	AdaC3	C = 0.8	0.9264	0.9221	0.8768	0.9570	0.9152	0.9305 ± 0.0148	0.9309 ± 0.0147

Bold values mean the best result in each category.



(a) porosity defect (IR=10.23)
(b) trimming defect (IR=9.09)
(c) material defect (IR=15.33)
(d) imprint defect (IR=1.41)

Figure 7. Performance comparison among 8 BQC2 methods for four types of die-casting defects.

lots, which involved the imbalanced classification of predicting normal or defective lots based on the manufacturing process conditions. To solve this problem, the preprocessing of the manufacturing quality data was first introduced. Since process condition data are often time-series data that are gathered from machines and their environments, the data need to be associated with quality inspection data based on timestamps. Next, to assign binary classes (i.e. defect or normal labels) to the production lots, two kinds of defect rate-based labelling methods, absolute rate-based or relative rate-based labelling, were suggested.

To build the BQC2 models for the preprocessed data, the AdaC1, AdaC2 and AdaC3-based BQC2 methods were implemented, which embrace the AdaBoost ensemble approach but consider different penalties for the misclassifications of majority and minority classes. Using the die-casting process data set, it was showed that the proposed AdaC2-based BQC2 classifier outperformed the other classifiers in classification categories such as classic classifiers and ensembles, cost-sensitive single classifiers and sampling-based ensembles in terms of F1-measure, *G*-mean and AUC, which are the main performance evaluation measures for imbalanced data classification.

It is expected that the implemented algorithms can be used to predict whether certain manufacturing process conditions likely result in the production of a specific defect type based on historical data that have been taken from a real-world manufacturing environment. If such valuable data are collected and combined for the purposes of continuous improvement, data-driven manufacturing analysis techniques such as those proposed in this paper can be utilised to systematically improve the product quality and the operational efficiency in real-world manufacturing.

This study on data-driven manufacturing analysis can be improved in few ways. First of all, data source is crucial in this kind of data-driven analysis. A variety of sensors such as vibration and sound can be adopted to gather more precise production conditions for quality control. Many other preprocessing techniques such as time-series analysis and frequency analysis can be applied to extract more effective features from sensory data source. Finally, new recently developed algorithms for imbalanced classification including resampling-based and algorithm-based techniques can be considered to improve the performance of the QC2 solutions proposed in this research.

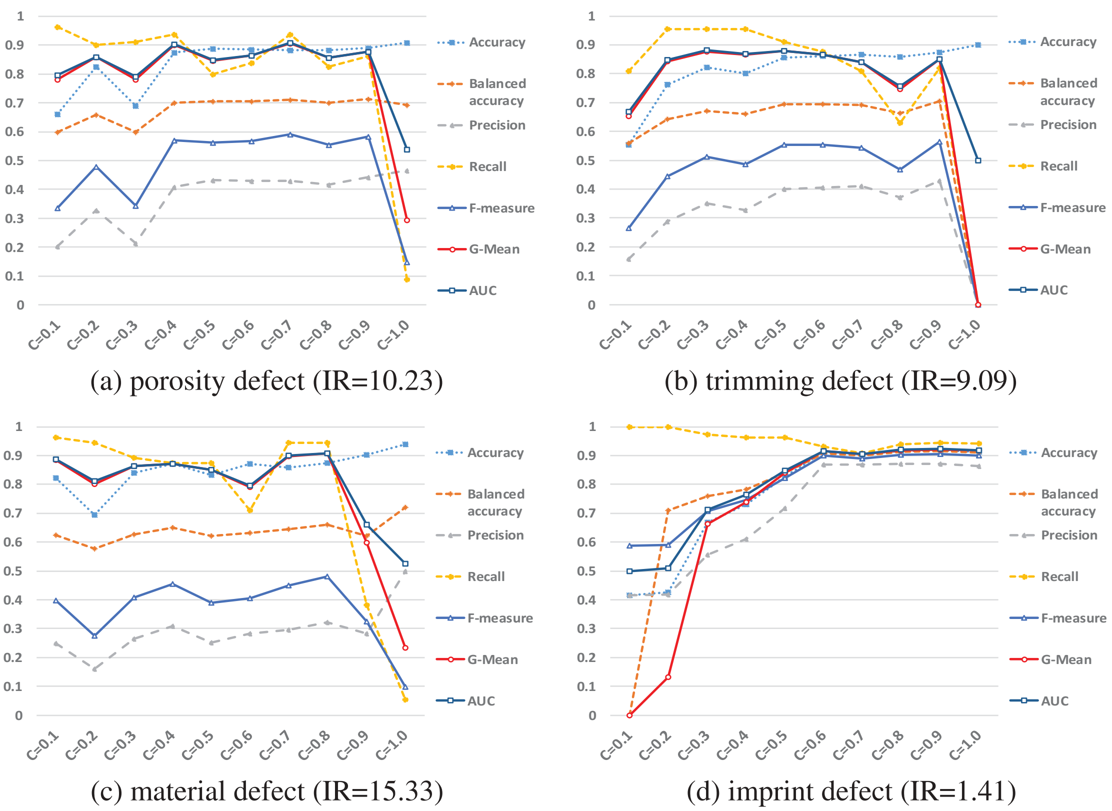


Figure 8. Performance changes of AdaC2-based BQC2 classifiers according to cost factor *C*. The best cost values were, respectively, (a) $C^* = 0.7$ for porosity defect, (b) $C^* = 0.5$ for trimming defect, (c) $C^* = 0.8$ for material defect, and (d) $C^* = 0.9$ for imprint defect.

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